

PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 7 Exam Notes: Microwave Antennas

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Antenna Parameters

Antennas are transitions between guided waves and free-space radiation. They are characterized by several fundamental parameters.

Effective Aperture (A_e): A parameter describing the power-collecting ability of an antenna. It is the ratio of received power to the incident wave power density:

$$A_e = \frac{P_{\text{rec}}}{W_i} \quad (\text{m}^2) \quad (1)$$

Core Antenna Parameters

1. Directive Gain (G_d) & Directivity (D): Directivity is the maximum value of directive gain in the peak radiation direction.

$$D = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}} \quad (2)$$

2. Aperture Gain Relation:

$$G = D \cdot \eta_{\text{rad}} = \frac{4\pi A_e}{\lambda^2} \quad (3)$$

3. Beamwidth (HPBW & FNBW): Half-Power Beam Width (HPBW) is the angular span between the 3 dB down points. First-Null Beam Width (FNBW) is the span between the first zero-intensity nulls.

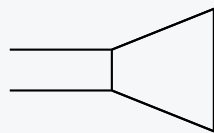
$$\text{FNBW} \approx 2 \times \text{HPBW} \quad (4)$$

2. Ground-Based Systems

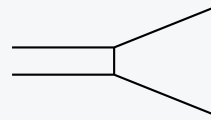
2.1 Horn Antenna

Horn antennas flare the end of a waveguide to match the waveguide impedance to the intrinsic impedance of free space (377Ω), minimizing reflection.

Horn Antenna Types



Pyramidal Horn



Sectoral Horn

Pyramidal Horn Gain:

$$G = \frac{4\pi}{\lambda^2} \epsilon_{ap} A = \frac{4\pi}{\lambda^2} \epsilon_{ap} (a_e b_e) \quad (5)$$

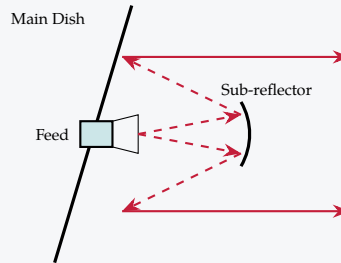
where a_e, b_e are the aperture widths and ϵ_{ap} is the aperture efficiency.

2.2 Parabolic Reflector Antenna

Reflector antennas focus a feed horn's spherical wavefronts into highly directive plane waves.

Cassegrain Feed: Uses a primary parabolic reflector, a secondary hyperbolic sub-reflector at the focus, and a feed horn placed at the vertex of the primary dish.

Cassegrain Feed Ray-Tracing Geometry



Gain: $G = \left(\frac{\pi D}{\lambda}\right)^2 \eta_{ap}$ where D is the diameter of the dish.

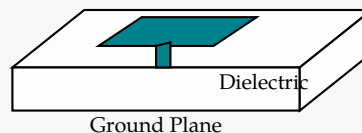
3. Airborne and Planar Antennas

Airborne antennas must be aerodynamic, low-weight, and often conformal (matching the contour of the aircraft skin).

3.1 Microstrip Patch Antenna

A microstrip patch antenna consists of a radiating metallic patch printed on one side of a thin dielectric substrate with a continuous ground plane on the other.

Microstrip Patch Antenna Structure



Feed Techniques:

1. **Microstrip Line Feed:** Easy to print, easy to match using inset cuts.
2. **Coaxial Probe Feed:** Feed probe is connected through a hole from the bottom ground, matching is done by adjusting the insertion point.
3. **Aperture-Coupled Feed:** Non-contact feed where a slot in the ground plane couples energy from a bottom line.
4. **Proximity-Coupled Feed:** Multi-layer line printed directly below the patch.

Quick Revision — Module VII

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Key Formulas at a Glance

Effective Aperture	$A_e = \frac{\lambda^2}{4\pi} G$
Horn Antenna Gain	$G = \frac{4\pi}{\lambda^2} \epsilon_{ap} (a_e b_e)$
Parabolic Dish Gain	$G = \left(\frac{\pi D}{\lambda}\right)^2 \eta_{ap}$
Parabolic HPBW	$\text{HPBW} \approx 70 \frac{\lambda}{D} \quad (\text{degrees})$
Patch Length	$L \approx \frac{c}{2f_0 \sqrt{\epsilon_{\text{eff}}}} - 2\Delta L$

One-Line Definitions

- **Directivity:** The ratio of maximum radiation intensity to average radiation intensity.
- **Pyramidal Horn:** A waveguide flaring in both electric and magnetic field planes to match free space.
- **Cassegrain Feed:** A dual-reflector antenna utilizing a parabolic dish and a hyperbolic sub-reflector.
- **Aperture Efficiency:** The ratio of the effective antenna aperture to its physical geometric area.
- **Polarization:** The orientation of the electric field vector of the radiated wave as it propagates.
- **Conformal Antenna:** An antenna designed to follow the curved contour of an host aircraft or vehicle.
- **Aperture Coupling:** A non-contact microstrip feeding technique using a slot in a shared ground plane.

Frequently Asked / Viva Questions

Q1. State the relation between antenna gain, directivity, and efficiency.

$G = \eta_{\text{rad}} \cdot D$, where η_{rad} is the radiation efficiency. For a lossless antenna, gain equals directivity ($G = D$).

Q2. What are the advantages of Cassegrain feeds over focal-point feeds?

Shorter waveguide feed lines (reducing insertion loss), lower spillover noise from the ground, and easier access to the feed horn at the vertex.

Q3. Explain the physical operation of a microstrip patch antenna.

The patch acts as a resonant cavity. Radiation occurs primarily from the fringing fields at the open-circuited edges of the patch.

Frequently Asked / Viva Questions (contd.)**Q4. Why does flaring a waveguide into a horn antenna increase directivity?**

Flaring increases the physical aperture area of the waveguide. Since $D \propto A/\lambda^2$, a larger aperture yields higher directivity and narrower beamwidth.

Q5. What is circular polarization and when is it preferred?

A wave polarization where the E-field vector rotates in a circle. It is preferred in satellite links because it is immune to Faraday rotation in the ionosphere.

Q6. Why do planar antennas require low-profile feed techniques in airborne systems?

To maintain aerodynamic properties, preventing wind drag, turbulence, and structural damage to the aircraft.

Q7. State the difference between E-plane and H-plane sectoral horns.

An E-plane horn flares only in the direction of the electric field. An H-plane horn flares only in the direction of the magnetic field.

Q8. How does the dielectric substrate thickness affect microstrip patch performance?

Thicker substrates increase bandwidth and radiation efficiency but lead to surface-wave excitation, causing radiation pattern degradation.

Q9. What is FNBW and how is it related to HPBW?

FNBW is First-Null Beamwidth. For symmetric main lobes, FNBW is approximately twice the HPBW ($\text{FNBW} \approx 2 \times \text{HPBW}$).

Q10. What is the purpose of the sub-reflector in a Cassegrain antenna?

To reflect the feed signals onto the main dish, allowing the feed electronics to be placed behind the main dish rather than suspended at the focus.

Q11. How do lens antennas work at microwave frequencies?

They use dielectric materials to collimate spherical wavefronts into plane waves, similar to optical lenses.

Q12. What is the typical gain range of a standard horn antenna?

Typically between 10 dB and 25 dB, depending on flaring and operating frequency.

Comparison of Microwave Antennas

Antenna Type	Typical Gain	Bandwidth	Physical Profile
Horn Antenna	Moderate (12 – 25 dBi)	Wide (waveguide range)	Bulky flared structure
Parabolic Dish	Very High (30 – 50 dBi)	Wide	Very bulky reflector assembly
Microstrip Patch	Low (5 – 8 dBi)	Very Narrow (2% – 5%)	Flat, lightweight, conformal
Lens Antenna	High (20 – 30 dBi)	Moderate	Heavy dielectric component